

Feature Specification

Student OS Documentation — File 06

Feature Template

Every feature must define purpose, user role, inputs, outputs, validation, permissions, states, errors, dependencies, edge cases and acceptance criteria.

Tasks

Input: title, subject, due date/time, priority, category, description, attachment, reminder.

Output: persisted task plus updated dashboard/calendar/notification state.

Validation: title required; date/time valid; priority/category from allowed values.

Edge cases: overdue task, completed task edited after completion, deleted subject.

Timetable

Input: subject, weekday, start/end, room, faculty, type.

Output: schedule record and related Today/Calendar views.

Validation: valid time range; configurable overlap warning/prevention.

Attendance

Input: attended and total classes or class events.

Output: percentage, target comparison, warning state and chart.

Formula: $\text{percentage} = \text{attended} / \text{total} \times 100$ when $\text{total} > 0$.

Never show misleading percentage when total is zero.

Exams

Input: title/subject, date/time, room, syllabus, preparation progress.

Output: countdown, calendar event, dashboard item and reminders.

Notes

Input: title, subject, content, attachments/links.

Output: searchable note.

Future AI actions must use only user-authorized content.

Notifications

Each notification has type, title, message, timestamp, read state and optional related entity.

Notifications should deep-link to the relevant feature when possible.

AI Features

All AI features require explicit inputs, clear output boundaries, failure handling, provider abstraction and user-visible indication that content is AI-generated.